Artificial Intelligence

VICTORIA UNIVERSITY OF WELLINGTON
TE HERENGA WAKA

AIML320/420

Course Overview

Dr Fangfang Zhang

SECS & Centre for Data Science and AI

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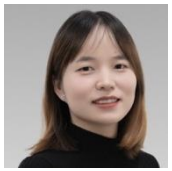
Outline

- Who and where
- Lectures/Discussions/Helpdesks
- Course focus and objectives
- Course materials
- Assessments and workload
- Policies
- Use of AI

Our team: Lecturers

- **Course Coordinator**
  - Dr Fangfang Zhang (fangfang.zhang@vuw.ac.nz, CO340)



  Need to meet with Fangfang? Book a time here

- **Lecturers**
  - Dr Heitor Murilo Gomes (heitor.gomes@vuw.ac.nz)
  - Prof Ali Knott (ali.knott@vuw.ac.nz)

- 3 tutors: Yuan Tian,
  Zhicheng Wu,
  Luyao Zhu

- 65 students (26 for AIML320 + 39 for AIML420)
- Can contact us before/after class, or email

Our team: Class Representatives

- A class rep is the **bridge** between the lecturers and the students, facilitate feedback by communicating regularly with the class and the course coordinator.
- i.e. somebody to raise issues, comments, suggestions with us on your behalf!
- Email me before Friday (27th Feb 2026) with brief message.

Our team: Class Representatives

AIML 420 (2026) - Home Page

Welcome to the homepage for **AIML 420** in 2026.

This course addresses concepts and techniques of artificial intelligence (AI). It provides a brief overview of AI history and search techniques, as well as covering important machine learning topics, tools, and algorithms with their applications, including neural networks and evolutionary algorithms. Other topics include analysing data, probability and Bayesian networks, planning and scheduling. The course will also give a brief overview of a selection of other current topics in AI.

Time and Location:

- Tuesday: 12:00 - 12:50 – LT220, Murphy Building, Kelburn
- Wednesday: 12:00 - 12:50 – LT220, Murphy Building, Kelburn
- Thursday: 12:00 - 12:50 – LT118, Laby Building, Kelburn

Helpdesk:

- Every **Tuesday** and **Thursday** from **2pm** to **3pm** at **CO242B** (week 2 to week 12, excluding teaching break).

People:

- Fangfang Zhang: fangfang.zhang@vuw.ac.nz, Room CO340 (coordinator and lecturer)
- Heitor Gomes: heitor.gomes@vuw.ac.nz, Room CO341 (lecturer)
- Ali Knott: ali.knott@vuw.ac.nz, Room AM226 (lecturer)
- Tutor: Yuan Tian, Zhicheng Wu, Luyao Zhu
- Class rep: TBD

Lectures, Discussions, Helpdesks

- **Lectures:**
  - **Tuesday** 12:00-12:50 (LT220, Murphy Building, Kelburn)
  - **Wednesday** 12:00-12:50 (LT220, Murphy Building, Kelburn)
  - You can ask questions at any time!
- **Tutorials:**
  - **Thursday** 12:00-12:50 (LT118, Laby Building, Kelburn)

In person, and all lectures will be recorded

- **Lectures**: Concepts, theories, ideas, methods
- **Tutorials**: Review of the concepts and theories, discussions, more thinking for students

Lectures, Discussions, Helpdesks

- **Helpdesks**:
  - **Tuesday** 2-3pm (CO242B, Cotton Building, Kelburn)
  - **Thursday** 2-3pm (CO242B, Cotton Building, Kelburn)
- Helpdesk starts from week 2 to week 12
- **Helpdesks are in-person**
- No helpdesks during teaching break
- No helpdesks during public holidays
- Get to know each other
- Special cases will be handled case by case
  - Mid-term test and final exam will be in person only

Course Objectives

- Understand and apply fundamental concepts and techniques of Artificial Intelligence
- Form a basis for further learning and research in AI
  - part of **AI Specialization** in **COMP** and **SWEN** majors
  - prerequisite of 400 AI courses (AIML 425, 426, 428, partly 427, 429)
- Know the working process of classical approaches
- Present you work in writing

Course Outline

Week	Num	Lecturer	Topic
1	1	Fangfang	Course Overview
^	2	Fangfang	Machine Learning Fundamentals
^	3	Fangfang	Tutorial 1
2	4	Fangfang	Supervised learning: K-Nearest Neighbours
^	5	Fangfang	Supervised Learning: Decision Trees
^	6	Fangfang	Tutorial 2
3	7	Fangfang	Supervised learning: Ensemble Learning
^	8	Fangfang	Data Preparation and Feature Engineering 1
^	9	Fangfang	Tutorial 3
4	10	Heitor	Data Preparation and Feature Engineering 2
^	11	Heitor	Data Preparation and Feature Engineering 3
^	12	Heitor	Tutorial 4
5	13	Heitor	Unsupervised Learning: Clustering
^	14	Heitor	Unsupervised Learning: Anomaly Detection
^	15	Heitor	Tutorial 5
6	16	Heitor	Probability Basics
^	17	Heitor	Bayes Theorem and Naive Bayes Algorithm
^	18	Heitor	Tutorial 6: unsupervised learning, probability theory and NB

Trimester break

Course Outline

7	19	Fangfang	Search & Optimisation (Search Space, Hill climbing, Simulated Annealing)
^	20	Fangfang	Evolutionary Computation and GA
^	21	Fangfang	Mid-term test
8	22	Fangfang	Genetic programming: Basics, Regression and Classification
^	23	Fangfang	Genetic programming: Automatic Heuristic Design
^	24	Fangfang	Tutorial 7
9	25	Fangfang	Planning and Scheduling
^	26	Fangfang	Planning and Scheduling
^	27	Fangfang	Tutorial 8
10	28	Ali	Neural Networks 1
^	29	Ali	Neural Networks 2
^	30	Ali	Tutorial 9
11	31	Ali	Neural Networks 3
^	32	Ali	Neural Networks 4
^	33	Ali	Tutorial 10
12	34	Ali	Other topics 1
^	35	Ali	Other topics 2
^	36	Ali	Course Review

Course Materials

- No text book. Check library and Google (Scholar)
- Stuart J. Russell and Peter Norvig, "Artificial Intelligence: A Modern Approach"
  - Not required, but useful especially if you want to study other topics
  - There are a lot of materials available online
- No background at all, you can read the book:

  Oliver Theobald, "Machine Learning For Absolute Beginners"
- Check the course outline on the website for more details
  - https://ecs.wgtn.ac.nz/Courses/AIML320_2026T1/CourseOutline
  - https://ecs.wgtn.ac.nz/Courses/AIML420_2026T1/CourseOutline

Course Materials

- **Announcements**
  - Nuku
  - Weekly email

- You can raise questions in Nuku Discussions

- These materials will be updated from time to time

Assessments

- **Assignments and Tests (60% in total)**
  - **Assignment 1 (15%)**: due by 26th March (Thursday), week 5
  - **Mid-term test (15%)**: on 23rd April (Thursday), week 7
  - **Assignment 2 (15%)**: due by 30th April (Thursday), week 8
  - **Assignment 3 (15%)**: due by 28th May (Thursday), week 12
- All assignments are due by **23:59pm**
- **Final Test (40%): closed-book, 2 hours**

- **3 late days:** Automatically applied, don't need to ask
- Send us request for asking extensions if you have special circumstances
  - One grade deduction per day late (5 marks), 0 marks after 3 days, and will not be marked beyond 7 days after the deadline.

Workload

- **10 hours** per week
  - Lectures: 2 hours
  - Reading, reviewing and doing assignments: 8 hours
- More time might be required if you didn’t adhere to all the prerequisites
- **Assumptions:**
  - Interested in AI
  - Abilities to find things by yourself!

Rules and Policies

- Plagiarism is **NOT tolerated**
  - Submit someone else's work as your own, including
    - material from a published source such as a library book, a journal article, etc.
    - material from an on-line software library, web pages, etc.
    - the work of another student, friend, relative, etc.
- Penalties:
  - zero marks for the work to which it relates
  - greater penalties in accordance with the University's Statute on Student Conduct

- If you discussed with other students, state it clearly in the report to avoid plagiarism issue

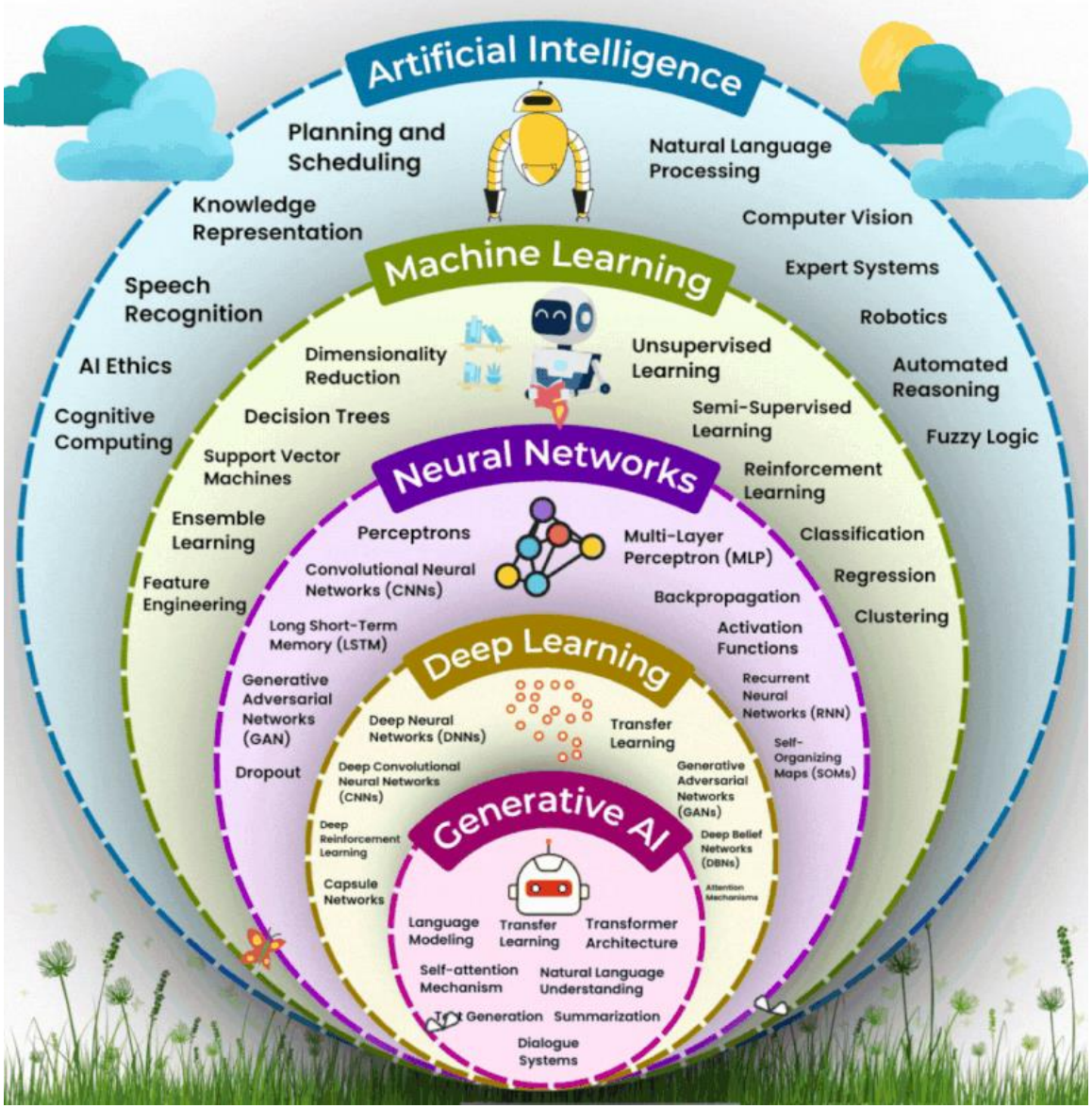
- https://ecs.victoria.ac.nz/Main/ComputerSciencePlagiarism

Use of AI

- **AI Green:** You are encouraged to use AI tools to assist in the learning and production of content. Students need to reference the technology they are using and will be assessed assuming full access to AI tools.
- Please **cite** whatever AI tool you have used, and how you have used them for your assignments
  - E.g., I have used Copilot to produce the code for the genetic algorithm for scheduling, …
  - I have used ChatGPT to get the answer to the question on …

- **NB**: The assignments will be mainly on **your understanding** of the concepts and algorithms. Thus, the discussions will be very important (worth many marks) in the report.
- **NB**: The final exam will be close-book, so you cannot use AI tools for the exam.
- **Consider this**: If you have to complete a task in an exam without using AI, would you be okay?

The AI Universe



What is a good Report?

1. Answer the question **directly** (i.e., marking is based on key points) --- good to start with a summary sentence and then explain details
2. Answer/give **what are asked** ("useless information" is not helpful)
3. **Show your thinking** rather than "copying"!!! (e.g., depends…)
4. Proper font size (e.g., clear)
5. Not necessary to be very long (keep concise and accurate)

What is a good Report?

6. Show your **working process**
7. **Do not bury your effort** (e.g., improvements of assignments)
8. Visualisation (e.g., Figures, tables)
9. **Bullet points** to make your report easier to read
10. No ideas about the assignments at all (e.g., show your thinking)

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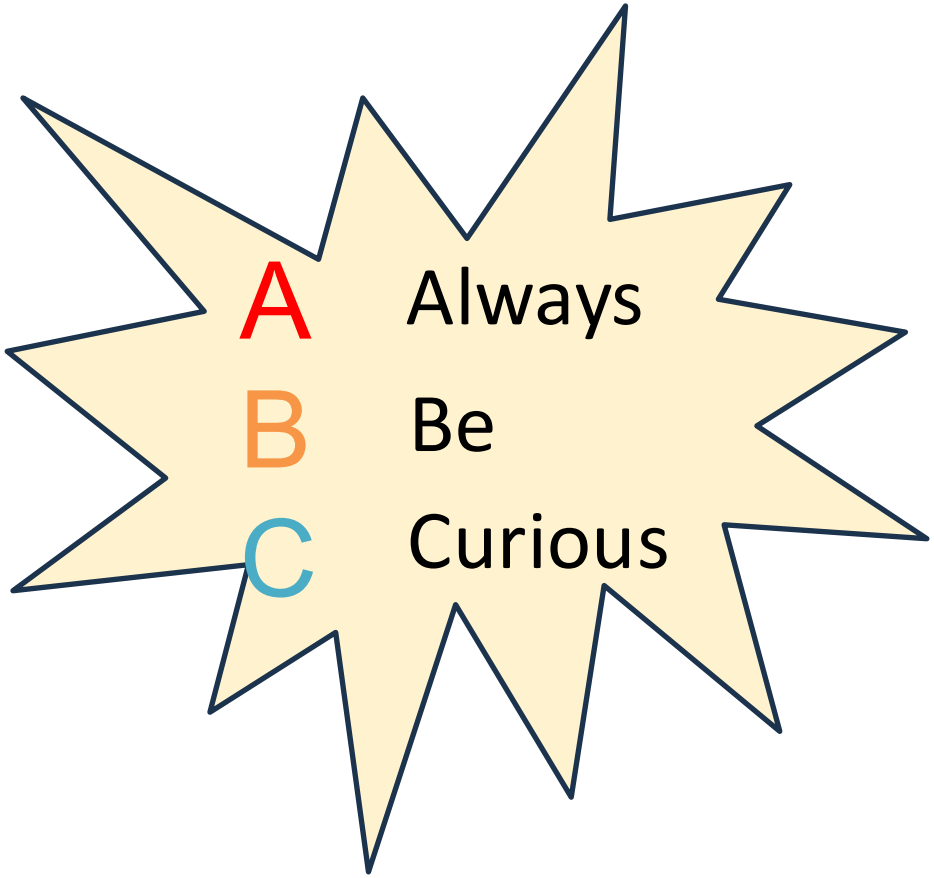
Secret for Learning